Listing 7.33 Specifying index/search analyzers during index creation

```
PUT authors_index_with_both_analyzers_field_level
{
  "mappings": {
    "properties": {
      "author_name":{
        "type": "text",
        "analyzer": "standard",
        "search_analyzer": "simple"
      }
    }
  }
}
```

The `author_name` field uses a `standard` analyzer for indexing and a `simple` analyzer during search.

ORDER OF PRECEDENCE

The engine begins to use the analyzers based on the following order of precedence when it finds analyzers at various levels (highest precedence first):

1. An analyzer defined at a query level
2. An analyzer defined by setting the `search_analyzer` property on a field when defining the index mappings
3. An analyzer defined at the index level
4. The indexing analyzer set on a field or an index

Now that we understand the built-in analyzers and how to create custom analyzers, it is time to look into the individual components of analyzers in detail. In the next three sections, we review the components that form an analyzer: tokenizers and character and token filters. Let's start with character filters.

7.6 Character filters

When a user searches for answers, the expectation is that they won't search with punctuation or special characters. For example, there is a high chance a user may search for "cant find my keys" (without punctuation) rather than "can't find my keys !!!". Similarly, the user should not search for the string "<h1>Where is my cheese?</h1>" (with the HTML tags). We also don't expect the user to search using XML tags like "<operation>callMe</operation>". Search criteria don't need to be polluted with unneeded characters. We also don't want users to search using symbols: ? instead of "alpha" or ? in place of "beta", for example.

Based on these assumptions, we can analyze and clean the incoming text using character filters. Character filters help purge unwanted characters from the input stream. Although they are optional, if they are used, they form the first component in the analyzer module.